Kshama Purohit

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PROFESSIONAL SUMMARY

Analytics-driven individual with a 4.0 GPA and dual degrees in Business Analytics and Computer Science, combining hands-on consulting project experience with deep technical expertise in data science, machine learning, and NLP. Proven ability to translate complex data into strategic, investment-ready recommendations for business stakeholders.

EDUCATION

M.Sc. Business Administration - Global Business Analytics | Virginia Tech *Aug 2025 – May 2026*
CGPA: 4.0/4.0 | Student Manager (VT Dining Services)

B.Sc. Business Information Technology - Cybersecurity Management & Analytics | Virginia Tech *Aug 2024 – May 2025*
CGPA: 4.0 / 4.0 | First Honours Graduate, Pamplin Senior Award (Academic Excellence)

B.Tech. Computer Science & Engineering - Data Science | NMIMS *Sept 2021 – May 2024*
CGPA: 3.6 / 4.0 | Joint Secretary (Social Impact), Vice President (Analytika), Board of Directors (RC Bombay Airport)

WORK EXPERIENCE

Graduate Teaching Assistant | Pamplin Engage Undergraduate Teaching Assistantship Program *Jan 2026 – May 2026*

- Advised and mentored 70+ students on data modelling, marketing analytics, and cybersecurity, translating complex technical concepts into structured, actionable guidance.
- Managed course operations via Canvas LMS maintaining 100% FERPA compliance while balancing 12-credit hours.

Data Science Intern | Transvolt Mobility Pvt. Ltd. *Jun 2025 – Aug 2025*

- Engineered an end-to-end analytics pipeline using Python and advanced ML models to forecast EV battery degradation from real-time CANBUS data, achieving 94% accuracy and deploying a production-grade cloud dashboard on AWS EC2 with executive-facing visuals surfacing fleet health KPIs and asset management insights.
- Translated model outputs directly into business outcomes: reduced reactive maintenance decisions, improved asset lifecycle planning, and delivered a fully operational ML system, from raw data to stakeholder-ready insight, supporting data-driven cost recommendations for leadership.

Undergraduate Research Assistant | VT Center for Excellence in Learning and Teaching *Mar 2025 – May 2025*

- Conducted structured data collection and web-scraping on AI adoption trends in analytics education, to synthesize findings into actionable curriculum recommendations for cross-functional academic stakeholders.

Data Science & Business Analysis Intern | The Sparks Foundation *May 2023 – Jun 2023*

- Validated student performance datasets to identify correlations between study habits and academic outcomes; built a forecasting model, translating analytical findings into structured, decision-ready recommendations for executive teams.

DATA ANALYTICS PROJECTS

Commercial Due Diligence & Growth Strategy: HelloFresh | Green Chef

- Processed 10,000+ raw customer comments using Python NLP and manual coding frameworks to identify churn drivers and unmet need states, translating unstructured data into structured, decision-ready business recommendations.
- Diagnosed a 55% subscription pause rate and 7% cancellation rate; iteratively developed a product solution, projecting $38M+ profit after taxes, with retention improvements of 6+ weeks per customer, effectively delivering investment-ready recommendations to business stakeholders.

Strategic Analysis of India's Carbon Footprint: Correlating Energy Consumption and Emissions Trends

- Ranked 17th globally in the SAS Curiosity Cup; synthesized large-scale ESG datasets using SAS Visual Analytics to identify India's key emission drivers and deliver actionable visualizations supporting strategic environmental forecasting.

VT Infrastructure Stress Test: Tableau & Power BI Dashboards

- Used survey data to model and stress-test Virginia Tech infrastructure capacity; built interactive dashboards and narrative data stories in both Tableau and Power BI, translating raw survey inputs into decision-ready visual intelligence to aid the planning committee in infrastructure-related decision making.

Customer Acquisition Strategy: Amazon Customer Churn Prediction & Analysis

- Built a multi-model churn prediction system using supervised and unsupervised machine learning, including K-means and Hierarchical Clustering for customer segmentation, producing a data-to-decision pipeline visualized in JMP.

SKILLS, CERTIFICATIONS & INTERESTS

Data & Analytics: MS Excel (Advanced; VBA, VLOOKUP), Power BI, Tableau, SAS Visual Analytics, SQL, Python.
Finance & Strategy: Business Requirements & Customer Insight Analysis, Market Sizing, Agile Project Development, Competitive Landscape Analysis, Stakeholder Communication, Process Improvement, Financial Modelling, Variance Analysis**.**
Cloud & Engineering: AWS EC2, Docker, Python (ML/Predictive Modelling), Data Pre-processing, Nginx, Gunicorn.
Certifications: AI - Product Management (Pendo, 2026), MySQL - Data Analytics and BI (Udemy, 2024), Student Marketing Certification (ANA, 2026).
Interests: Animal welfare, Painting, Baking, Reading, Stock trading

COMMUNITY INVOLVEMENT

Volunteer teacher (Spark-A-Change); Organized camps, drives, welfare campaigns that impacted 200+ community members.